I. Exercise 9.5 (Prove Theorem 9.4.)

Actually we want to prove the following two inequalities:

$$\frac{||\mathbf{r}||_2 \cdot ||\mathbf{x}||_2}{||A||_2 \cdot ||A^{-1}||_2 \cdot ||\mathbf{b}||_2 \cdot ||\mathbf{e}||_2} \leq 1 \quad \text{and} \quad \frac{||\mathbf{b}||_2 \cdot ||\mathbf{e}||_2}{||A||_2 \cdot ||A^{-1}||_2 \cdot ||\mathbf{r}||_2 \cdot ||\mathbf{x}||_2} \leq 1 \tag{1}$$

We have $||A^{-1}||_2 \cdot ||\mathbf{b}|| \geq ||A^{-1}\mathbf{b}||_2 = ||\mathbf{x}||_2$ and $||A||_2 \cdot ||\mathbf{e}|| \geq ||A\mathbf{e}||_2 = ||\mathbf{r}||_2$. Hence for the first inequality:

$$\frac{||\mathbf{r}||_2 \cdot ||\mathbf{x}||_2}{||A||_2 \cdot ||A^{-1}||_2 \cdot ||\mathbf{b}||_2 \cdot ||\mathbf{e}||_2} \leq \frac{||\mathbf{r}||_2 \cdot ||\mathbf{x}||_2}{||\mathbf{x}||_2 \cdot ||\mathbf{r}||_2} = 1$$

Also, $||A^{-1}||_2 \cdot ||\mathbf{r}|| \geq ||A^{-1}\mathbf{r}||_2 = ||\mathbf{e}||_2$ and $||A||_2 \cdot ||\mathbf{x}|| \geq ||A\mathbf{x}||_2 = ||\mathbf{b}||_2$. Hence for the second inequality:

$$\frac{||\mathbf{b}||_2 \cdot ||\mathbf{e}||_2}{||A||_2 \cdot ||A^{-1}||_2 \cdot ||\mathbf{r}||_2 \cdot ||\mathbf{x}||_2} \leq \frac{||\mathbf{b}||_2 \cdot ||\mathbf{e}||_2}{||\mathbf{e}||_2 \cdot ||\mathbf{b}||_2} = 1$$

Hence Theorem 9.4 is proved.

II. Exercise 9.8 (Compute cond(A) for $n = 8$ and $n = 1024$.)

We konw $\lambda_k(A) = 4n^2 \sin^2 \frac{k\pi}{2n}$ for $k = 1, 2, ..., n-1$. Since A is symmetric, we got $||A||_2$ immediately by

$$||A||_2 = \max \sqrt{\lambda(A^T A)} = \max |\lambda(A)| = 4n^2 \sin^2 \frac{(n-1)\pi}{2n}.$$

Since $\lambda_k(A^{-1}) = \lambda_k(A)^{-1}$, we can also got $||A^{-1}||_2$ by

$$||A^{-1}||_2 = \min |\lambda(A)^{-1}| = \frac{1}{4n^2 \sin^2 \frac{\pi}{2n}}.$$

It follows that

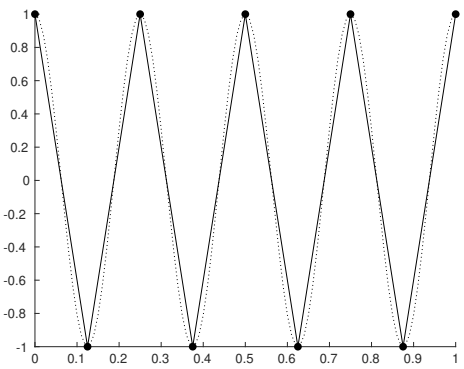
$$\text{cond}(A) = ||A||_2 \cdot ||A^{-1}||_2 = \frac{\sin^2 \frac{(n-1)\pi}{2n}}{\sin^2 \frac{\pi}{2n}}.$$

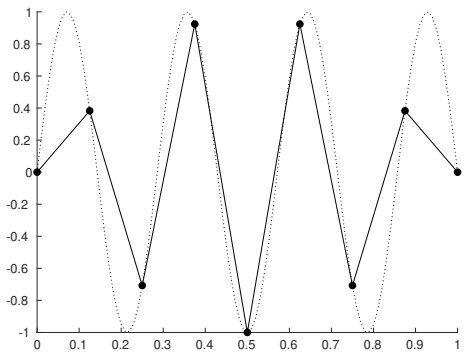
Use a calculator to compute the value for $n = 8$ and $n = 1024$, we got

$$\text{cond}(A)|_{n=8} \approx 48.99 \quad \text{and} \quad \text{cond}(A)|_{n=1024} \approx 1.046 \times 10^6.$$

III. Exercise 9.11 (Show the maximum wavenumber representable on Ω^h.)

$n_{\max}$ reaches when we alternate from local maximum and local minumum at the $n+1$ grid points. As the following figure shows.





$n_{\max} = \frac{1}{h}$

We can see $n_{\max} = \frac{1}{h}$. When we require the Fourier mode be 0 at the bondary points, the maximum number of alternation between local extrema is reduced by one. Hence in this condition, we have $n_{\max} = \frac{1}{h} - 1$.

IV. Exercise 9.14 (Plot the case of $n = 6$ for Example 9.13.)

As the following figure shows.

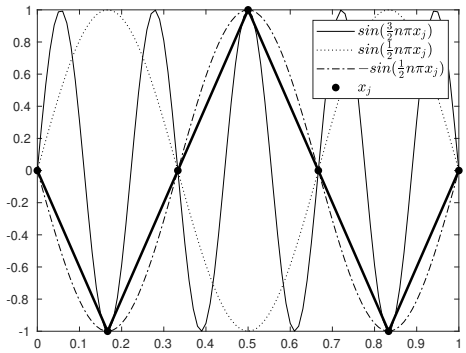


Figure 1: The case of $n = 6$ for Example 9.13.

V. Exercise 9.17 (Prove Lemma 9.16.)

For the kth eigenvalue λ_k and the corresponding eigenvector $\mathbf{w}_k$ we have:

$$(\lambda_k \mathbf{w}_k)_j = \left(1 - 2\omega \sin^2 \frac{k\pi}{2n}\right) \sin \frac{jk\pi}{n} = \left(1 - \omega + \omega \cos \frac{k\pi}{n}\right) \sin \frac{jk\pi}{n}$$

Now,

$$(T_\omega \mathbf{w}_k)_1 = (1-\omega) \sin \frac{k\pi}{n} + \frac{\omega}{2} \sin \frac{2k\pi}{n} = \left(1 - \omega + \omega \cos \frac{k\pi}{n}\right) \sin \frac{k\pi}{n} = (\lambda_k \mathbf{w}_k)_1.$$

And for each $2 \le j \le n-2$, we have

$$\begin{aligned}
(T_\omega \mathbf{w}_k)_j &= (1-\omega) \sin \frac{jk\pi}{n} + \frac{\omega}{2} \left(\sin \frac{(j-1)k\pi}{n} + \sin \frac{(j+1)k\pi}{n} \right) \\
&= (1-\omega) \sin \frac{jk\pi}{n} + \frac{\omega}{2} \left(\sin \frac{jk\pi}{n} \cos \frac{k\pi}{n} - \cos \frac{jk\pi}{n} \sin \frac{k\pi}{n} + \sin \frac{jk\pi}{n} \cos \frac{k\pi}{n} + \cos \frac{jk\pi}{n} \sin \frac{k\pi}{n} \right) \\
&= (1-\omega) \sin \frac{jk\pi}{n} + \omega \sin \frac{jk\pi}{n} \cos \frac{k\pi}{n} = (\lambda_k \mathbf{w}_k)_j.
\end{aligned}$$

Note that

$$\sin \frac{(n-1)k\pi}{n} \cos \frac{k\pi}{n} + \cos \frac{(n-1)k\pi}{n} \sin \frac{k\pi}{n} = \sin(k\pi) = 0$$

Then finally we have

$$\begin{aligned}
(T_\omega \mathbf{w}_k)_{n-1} &= (1-\omega) \sin \frac{(n-1)k\pi}{n} + \frac{\omega}{2} \sin \frac{(n-2)k\pi}{n} \\
&= (1-\omega) \sin \frac{(n-1)k\pi}{n} + \frac{\omega}{2} \left(\sin \frac{(n-1)k\pi}{n} \cos \frac{k\pi}{n} - \cos \frac{(n-1)k\pi}{n} \sin \frac{k\pi}{n} \right) \\
&= (1-\omega) \sin \frac{(n-1)k\pi}{n} + \frac{\omega}{2} \left(\sin \frac{(n-1)k\pi}{n} \cos \frac{k\pi}{n} + \sin \frac{(n-1)k\pi}{n} \cos \frac{k\pi}{n} \right) \\
&= (1-\omega) \sin \frac{(n-1)k\pi}{n} + \omega \sin \frac{(n-1)k\pi}{n} \cos \frac{k\pi}{n} = (\lambda_k \mathbf{w}_k)_{n-1}.
\end{aligned}$$

And hence $T_\omega \mathbf{w}_k = \lambda_k \mathbf{w}_k$. Then Lemma 9.16 is proved.

VI. Exercise 9.18 (Reproduce Fig. 2.7 and show slow convergence.)

Reproduce Fig. 2.7 in the book by Briggs et al.[2000] as following.

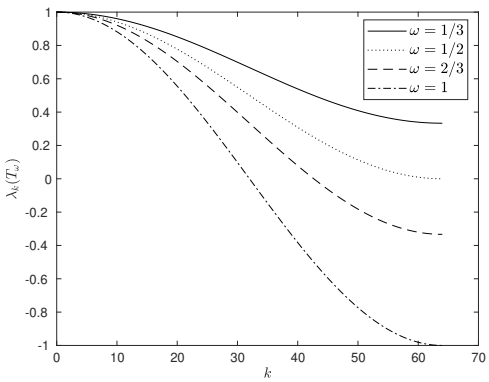


Figure 2: Reproduce Fig. 2.7 in the book by Briggs et al.[2000].

It was produced by the following **matlab** code.

```
w = [1/3, 1/2, 2/3, 1];
lstl = ['k', ':k', '--k', '-.k'];
ind = [1,1,2,3,4,6,7,9];
n = 64;
rho = zeros(1,4);
for wk = 1:4
    cw = w(wk);
    k = 0:0.1:n;
    lam = 1-2*cw*sin(k*pi/(2*n)).^2;
    rho(wk) = lam(11);
    plot(k,lam,lstl(ind(2*wk-1):ind(2*wk)))
    hold on
end
disp(rho);
xlabel('$k$','interpreter','latex');
ylabel('$\lambda_k(T_\omega)$','interpreter','latex');
legend('$\omega=1/3$','$\omega=1/2$','$\omega=2/3$','$\omega=1$','interpreter','latex',
'FontSize',12);
```

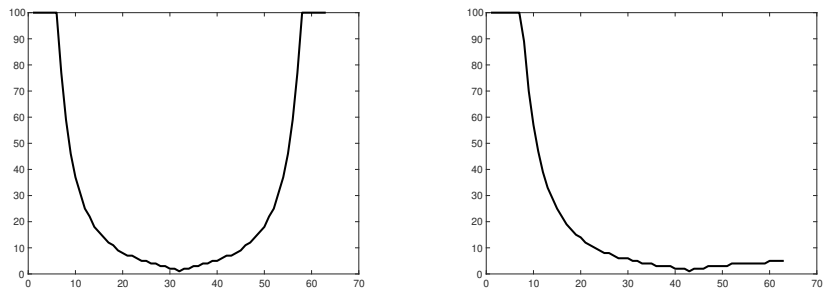
Clearly $\rho(T_{\omega_1}) > \rho(T_{\omega_2})$ for any $\omega_1 < \omega_2$. Then

$$\rho(T_\omega) \geq \rho(T_1) \approx 0.998795 > 0.9986 \quad \text{for all } \omega \in [0,1].$$

And hence slow convergence.

VII. Exercise 9.21 (Reproduce Fig. 2.8.)

Reproduce Fig. 2.8 in the book by Briggs et al.[2000] as following.



$\omega = 1$ (Regular Jacobi) $\omega = \frac{2}{3}$

Figure 3: Reproduce Fig. 2.8 in the book by Briggs et al.[2000].

We can see that the regular Jacobi is only good for damping modes $16 \le k \le 48$ while weighted Jacobi for $\omega = \frac{2}{3}$ is good for damping modes $16 \le k \le 64$.

The figures are produced by the following **matlab** code.

```
n = 64;
w = 2/3;
T = eye(n-1)*(1-w);
T(1:n-2,2:n-1) = T(1:n-2,2:n-1) + eye(n-2)*(w/2);
T(2:n-1,1:n-2) = T(2:n-1,1:n-2) + eye(n-2)*(w/2);
x = 1:n-1;
y = x;
for k = 1:n-1
    w = mode(n,k);
    wk = w;
    iter = 0;
    for i = 1:100
        w = T*w;
        iter = iter+1;
        if norm(w)/norm(wk)<0.01
            break;
        end
    end
    y(k) = iter;
end
plot(x,y,'k','LineWidth',2);
```

VIII. Exercise 9.35 (The computational cost of an FMG cycle.)

IN a D-dimesional domain with $n = 2^m$ cells ($m \in \mathbb{N}^+$) along each dimesion. The computational cost of an FMG cycle for $\nu_1 = \nu_2 = 1$ is

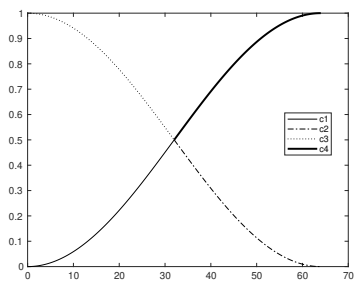
$$2\mathrm{WU}\sum_{j=0}^{m} j2^{-(j-1)D} \le 2\mathrm{WU}\sum_{j=0}^{\infty} j2^{-(j-1)D} = 2\mathrm{WU}\,\frac{\mathrm{d}}{\mathrm{d}x}\left(\sum_{j=0}^{\infty} x^j\right)\Bigg|_{x=2^{-D}} = \frac{2}{(1-2^{-D})^2}\mathrm{WU}$$

where WU is the computational cost of performing one relaxation sweep on the finest grid. When D= $1, 2, 3$, we have the corresponding upperbond 8WU, $\frac{32}{9}$WU and $\frac{128}{49}$WU.

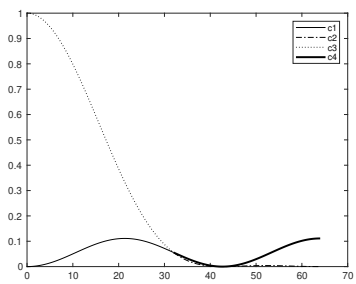
IX. Exercise 9.41 (Reproduce figures and estimate ρ(TG).)

For any $k \in [1, n-1]$, we always have $0 < \lambda_k, \lambda_{k'}, s_k, c_k < 1$. Hence when ν_1 and ν_2 are large enough, the magnitude of all four c_i's will be small.

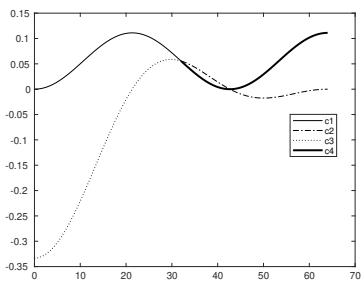
Here are the reproduced figures.



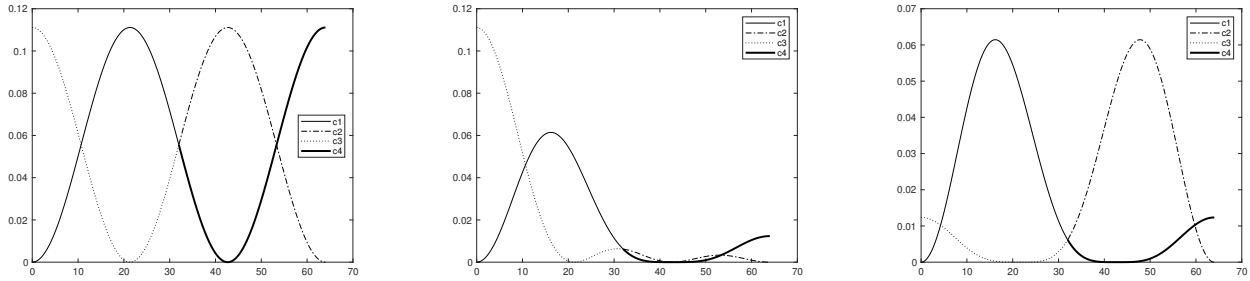
$\nu_1 = 0, \nu_2 = 0$



$\nu_1 = 0, \nu_2 = 2$



$\nu_1 = 1, \nu_2 = 1$



$\nu_1 = 2, \nu_2 = 0$ $\nu_1 = 2, \nu_2 = 2$ $\nu_1 = 4, \nu_2 = 0$

Figure 4: Reproduce figures in Exercise 9.41.

The figures are produced by the following **matlab** code.

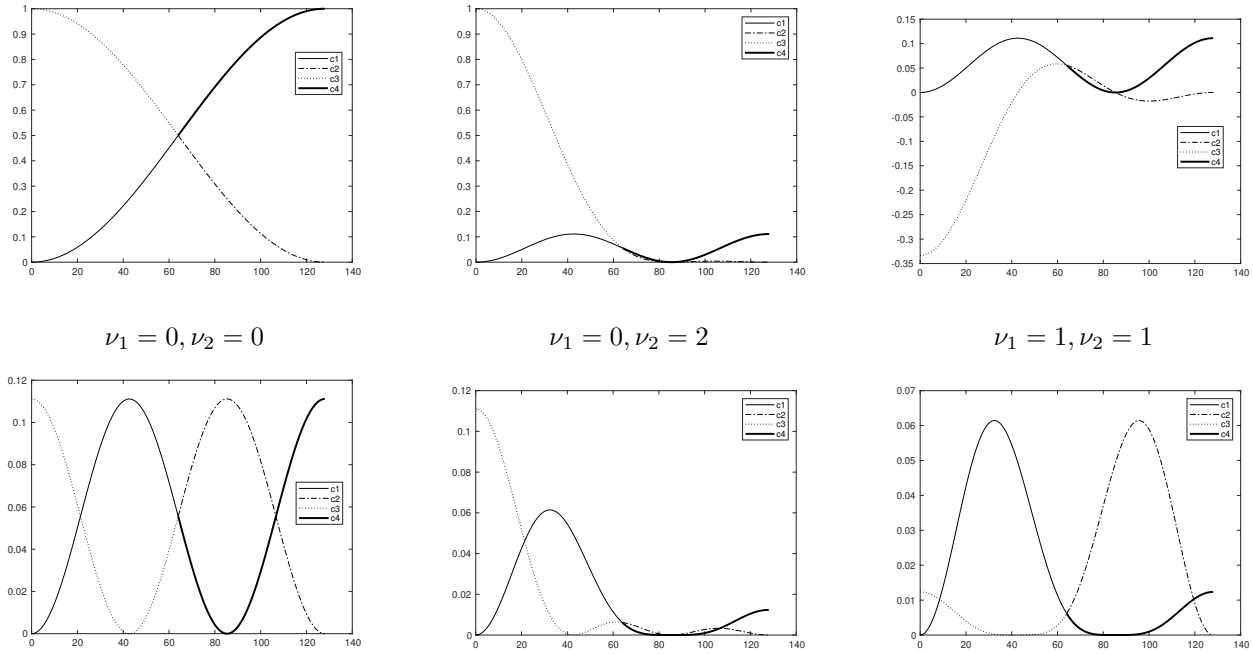
```
n = 64;
w = 2/3;
v1 = 2; v2 = 0;
lx1 = 0:0.1:n/2; lx2 = n/2:0.1:n;
y1 = lx1; y4 = lx2; y3 = lx1; y2 = lx2; rho = lx1;
for lk = 0:5*n
    ind1 = lk+1;
    ind2 = size(lx2,2)-lk;
    k = lk/10;
    lamk = lambda(n,k,w);
    lamk2 = lambda(n,n-k,w);
    sk = (sin(k*pi/(2*n)))^2;
    ck = (cos(k*pi/(2*n)))^2;
    y1(ind1) = lamk^(v1+v2)*sk;
    y4(ind2) = lamk2^(v1+v2)*ck;
    y3(ind1) = lamk2^v1*lamk^v2*ck;
    y2(ind2) = lamk^v1*lamk2^v2*sk;
    T = [y1(ind1),y2(ind2);y3(ind1),y4(ind2)];
    rho(ind1) = max(abs(eig(T)));
end
disp([min(rho),max(rho)]);
plot(lx1,y1,'k')
hold on
plot(lx2,y2,'-.k')
plot(lx1,y3,':k')
plot(lx2,y4,'k','LineWidth',2)
legend('c1','c2','c3','c4')

function lam = lambda(n,k,w)
    lam = 1-2*w*(sin(k*pi/(2*n)))^2;
end
```

The code will also print $\min \rho(\mathrm{TG})$ and $\max \rho(\mathrm{TG})$ for each ν_1, ν_2. Here are the results.

ν_1	ν_2	$\min \rho(\mathrm{TG})$	$\max \rho(\mathrm{TG})$
0	0	1	1
0	2	0.1111	0.1111
1	1	0.1111	0.1111
2	0	0.1111	0.1111
2	2	0.0123	0.0617
4	0	0.0123	0.0617

We can see that $\rho(\mathrm{TG}) \approx 0.1$ holds for $\nu_1 + \nu_2 = 2$. Moreover, we can reproduce the figures with $n = 128$ as following.



$\nu_1 = 0, \nu_2 = 0$ $\nu_1 = 0, \nu_2 = 2$ $\nu_1 = 1, \nu_2 = 1$

$\nu_1 = 2, \nu_2 = 0$ $\nu_1 = 2, \nu_2 = 2$ $\nu_1 = 4, \nu_2 = 0$

Figure 5: Reproduce figures in Exercise 9.41. for $n = 128$

We can see the result is independent of the grid size, which implies the independence of $\rho(\mathrm{TG})$ from the grid size.

X. Exercise 9.45 (Prove Lemma 9.44.)

By definition 9.25, the full-weighting operator $I_h^{2h} : \mathbb{R}^{n-1} \to \mathbb{R}^{\frac{n}{2}-1}$ is

$$I_h^{2h} = \frac{1}{4}\begin{bmatrix} 1 & 2 & 1 & & & & \\ & & 1 & 2 & 1 & & \\ & & & \ddots & \ddots & \ddots & \\ & & & & 1 & 2 & 1 \end{bmatrix},$$

which clearly has full row rank. Hence $\dim\mathcal{R}(I_h^{2h}) = \frac{n}{2} - 1$. By the property of linear map, we have

$$\dim\mathcal{N}(T) + \dim\mathcal{R}(T) = \dim(V) \quad \text{for any linear map } T : V \to W$$

Hence $\dim\mathcal{N}(I_h^{2h}) = n - 1 - \left(\frac{n}{2} - 1\right) = \frac{n}{2}$. That's proved.